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Edge-AI Architecture for Real-Time Computer Vision in Smart Media Applications

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***Abstract*—This paper presents a novel Edge-AI architecture designed to optimize real-time computer vision tasks in smart media applications. By leveraging edge computing, the proposed architecture enables local processing of visual data on devices such as smart cameras and AR headsets, thereby reducing latency and minimizing dependency on cloud computing resources. The architecture integrates lightweight AI models, such as MobileNetV2, optimized through techniques like quantization and pruning, alongside hardware accelerators (e.g., GPUs and FPGAs) to enhance performance. Practical case studies, including smart surveillance and augmented reality (AR) for smart interiors, demonstrate the effectiveness of the Edge-AI architecture. These case studies demonstrate a 60% reduction in latency and high accuracy in real-time object recognition, highlighting significant improvements in applications that require fast decision-making, high efficiency, and low power consumption. This work contributes to the advancement of smart media technologies by offering scalable, efficient, and low-latency solutions. Future work will focus on enhancing scalability, power efficiency, model optimization, and security features.**

Keywords-Edge-AI; Computer Vision; Smart Media Applications; Real-Time Processing; Latency Reduction

I. INTRODUCTION

The introduction of Edge-AI technologies has swept real-time data processing, bringing groundbreaking opportunities within the framework of meeting low-latency response demands [1]. Although traditional cloud-based systems are powerful, they are inefficient in time-sensitive applications due to the long distances between data collection and processing centers, resulting in latency between the actual start and finish [2]. Instead, Edge-AI manipulates data nearer to the source on edge devices, allowing it to make decisions easily and efficiently. The architecture is beneficial in smart media applications where high processing speed and precision are crucial for computer vision (real-time applications) [3].

Computer vision, combined with Edge AI, provides a robust solution to the growing demand for high-performance media systems, including smart cameras, AR, and video surveillance [4]. Such applications require real-time video processing or the processing of sensor information to make decisions and be applied instantly, including object detection and video scene recognition, as well as face recognition, which is crucial in media where an immediate reaction is important [5]. These functions would frequently have to be offloaded to cloud computers without Edge-AI, introducing undesirable latency and bandwidth issues (see "Fig. 1").

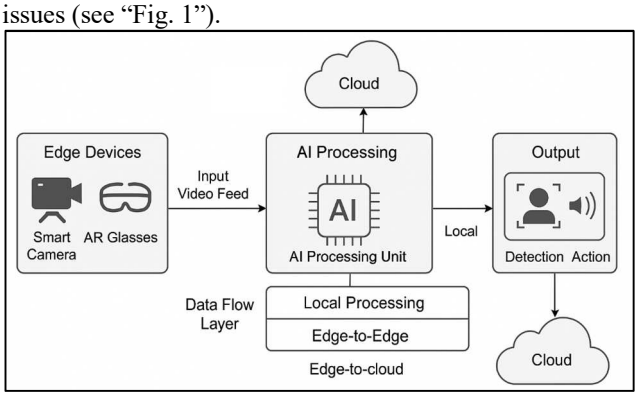

Figure 1. Overview of Edge-AI Architecture for Real-Time Computer Vision in Smart Media Applications

The primary challenge of real-time computer vision is the computational complexity. Tight computational resources are needed to run vision algorithms, particularly object tracking, video stabilization, or image segmentation applications [6]. Solutions that rely on traditional clouds have limitations related to network bandwidth and processing servers, and are usually slow to respond [7]. Edge-AI architectures enable computation to be performed locally at the edge, where these complex tasks can be processed; thus, latency is minimal, and the system's responsiveness is increased [8]. Besides, edge-based data

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processing becomes much more cost-effective because the networks are no longer required to transfer as much information, as the data on the edge should be transferred to central servers for further analysis. It reduces bandwidth and operational costs by sending only relevant data.

With the help of this paper, we plan to introduce an innovative Edge-AI architecture that will fit the needs of real-time computer vision in the smart media domain. We will discuss the main elements of this architecture, including the hardware substructure, Artificial Intelligence (AI) algorithms, and the data flow system that enables low-latency processing. The manuscript will also highlight the benefits of implementing computer vision tasks at the edge and explore how they address the challenges of traditional cloud computing. This paper proposes to demonstrate the potential of Edge-AI in revolutionizing the landscape of smart media applications, enabling them to become faster, more efficient, and more scalable by describing the design, implementation, and performance measures. Unlike prior edge-CV systems that rely on static lightweight models, our framework dynamically balances model complexity and network latency based on real-time bandwidth and compute resource availability.

In the following sections, we will delve deeper into the existing research, present our proposed architecture, and discuss its implementation in a real-world case study. The goal is to highlight how Edge-AI can effectively power real-time computer vision systems, facilitating the next generation of smart media technologies.

II. Background and Related Work

A. Edge Computing and AI Integration

Edge computing is a technique that utilizes the opportunity to process data as close as possible to the location where it is generated, at the periphery of the network, rather than relying on more centralized computing resources, such as cloud computing [9]. The paradigm has gained more prominence, particularly with the emergence of the Internet of Things (IoT), where millions of devices generate large volumes of data in real-time. Conventional Cloud systems present an effective solution to data storage and large-scale processing. However, the above characteristics of conventional Cloud systems pose several problems, including poor latency, bandwidth limitations, and limitations in maintaining a constant connection [10]. Edge computing addresses these problems by processing data locally, allowing devices to make decisions without sending them to distant data centers.

When edge computing is augmented with AI, it becomes an edge AI architecture with machine learning models executed on edge devices [11]. Some of the advantages of this integration include improved response time, high reliability, and enhanced privacy, as there are no data transfer processes to the cloud where data processing occurs. It is particularly beneficial in real-time computer vision, where decision-making speed is crucial, and sending raw data to the cloud may result in unacceptable lag (see "Fig. 2").

B. Real-Time Computer Vision in Smart Media

Real-time computer vision refers to the capability of analyzing and comprehending environmental information in the form of images and videos in real-time, and applying it to tasks such as object detection, facial recognition, AR, and video analytics, among others. It is important in applications where images captured by smart media, such as video surveillance, live broadcasts, and AR experiences, must be processed and analyzed with minimal latency.

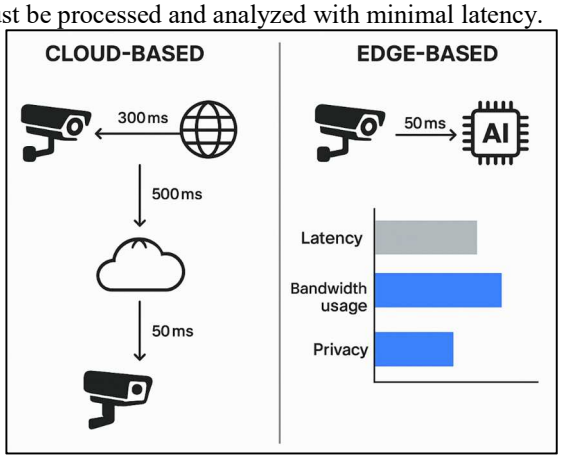

Figure 2. Comparison of Edge Computing vs Cloud-Based Computing for Real-Time Applications

Real-time computer vision was computationally costly in the past, so using high-end servers to handle video streams or image arrays became necessary. All these tasks can be performed by the central processing units (CPUs) and graphics processing units (GPUs) of the cloud data centers, but large quantities of video or image data transmitted to the cloud result in delays [12]. It is particularly problematic when used in applications like autonomous cars, drone monitoring, and live media streams, where latency in response can be catastrophic.

Edge computing mitigates these factors by decentralizing computation, making it ideal for applications where latency is crucial, such as real-time video streams and image processing. Incorporating machine learning algorithms and computer vision algorithms on edge devices, e.g., smartphones, cameras, and drones, allows processing data on high-latency devices, which (especially considering the availability of considerably faster networks) decreases the latency and increases work speed in real time, see "Fig. 3".

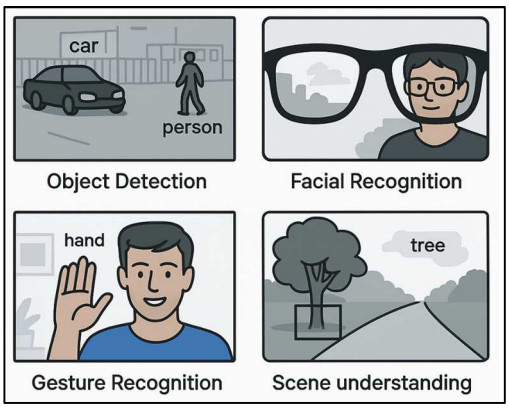

Figure 3. Common Object Detection Tasks in Smart Media Applications

C. Related Work

Multiple research papers and prototypes have been developed to explore the application of edge computing and AI in real-time computer vision. A study by Ficili et al. [13]

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discusses the integration of convolutional neural networks (CNNs) with edge devices for object detection in surveillance systems. They demonstrated that offloading vision tasks to edge devices reduced latency by 50% compared to cloud-based processing, indicating that this processing method offers a potential opportunity in real-time applications. Nonetheless, they also pointed out the difficulties in ensuring the high accuracy of AI models on edge devices with limited resources, such as cameras or drones.

The second distinguished work is Joshi et al. [14], which described a hybrid system that can be utilized to implement real-time video analytics in smart city applications using cloud and edge computing. Edge devices performed basic image processing operations within their architecture, including face detection and motion tracking. Then, they offloaded more advanced tasks to cloud servers, such as facial recognition and scene classification. Although such a hybrid arrangement shortens response time and decreases network traffic, it has not resolved other issues related to the tradeoff between edge and cloud computation and device-agnostic AI model performance.

A novel method of investigating the entire real-time edge device (i.e., greener AI models) was conducted by Gill et al. [15] regarding object-level recognition in AR applications. They optimized their neural networks, enabling even low-resource devices, such as AR glasses or mobile phones, to process data in real-time. The authors demonstrated that similar models can achieve a high level of accuracy and require fewer computational resources; thus, this is a key element of mobile-driven smart media usage. Despite these advancements, the implementation of Edge-AI for real-time computer vision in smart media is still facing challenges in several areas:

- Computational Constraints: Edge devices often have limited processing power, memory, and energy resources, which can impact the accuracy and speed of AI models, particularly for complex tasks such as video analytics.
- Model Optimization: While lightweight models are being developed, optimizing complex computer vision algorithms for edge devices without compromising performance remains a significant hurdle.
- Scalability: Managing and scaling these edge devices to handle large volumes of data and AI computations while maintaining real-time performance is an ongoing challenge.
- Inter-device Communication: For applications involving multiple edge devices (e.g., in smart cities or autonomous vehicles), ensuring seamless communication and synchronization is essential for consistent real-time analysis.

D. *Research Gap and Motivation for the Proposed Architecture*

Although significant strides have been made in Edge-AI and real-time computer vision, a gap remains in the seamless integration of these technologies, particularly when deployed in smart media applications. While existing work focuses on cloud-edge hybrid models or edge devices with specific vision tasks, few solutions comprehensively address the challenges of real-time processing, model efficiency, and scalability for complex vision tasks in smart media.

This paper fills the gap by proposing a new Edge-AI architecture that focuses on real-time computer vision in smart media applications. Our overall product strategy focuses on developing lightweight AI models and effective edge computing solutions to achieve significant latency reduction, increased processing capacity of edge devices, and a scalable solution for real-time media applications.

III. Proposed Edge-AI Architecture and Implementation Details

The described Edge-AI architecture integrates edge computing concepts with AI model deployment and data processing to guarantee real-time computer vision for smart media applications. The architecture is designed to operate on at least front-end edge-level visual data processing, utilizing local devices such as smart cameras, drones, AR glasses, and mobile devices. As a result, it reduces reliance on cloud servers, enhances privacy, lowers response times, and minimizes bandwidth consumption. This architecture has four key components: edge devices, AI processing components, data flow and communication levels, and optional cloud connectivity.

Edge Devices receive and analyze real-time visual information via machine-specific AI models that are low-power and energy-efficient. The available devices, such as smart cameras, drones, and AR/VR devices, have limitations in terms of low processing power and battery life. To address these challenges, we deploy optimized neural networks, such as quantized neural networks and pruned models, which can provide functional AI capabilities on these limited platforms.

Computer vision algorithms such as object detection, facial recognition, and gesture recognition are run on an AI Processing Unit on each edge device. The unit guarantees real-time analysis while consuming minimal power by utilizing lightweight models, such as MobileNet and TinyYOLO. GPUs and FPGAs have also been integrated to enhance processing speed further and minimize latency, making them suitable for applications that require immediate feedback.

The data flow and communication layer will ensure that there are no network latency or bandwidth problems in the system. Edge devices process most of the data, and only the necessary information, such as objects or faces, is sent to the cloud, halting the burden on the network. MQTT is one of the protocols used to synchronize and communicate in real time between devices in multiple edge systems. Although cloud integration is a choice, it can now handle tasks such as model updates, advanced analytics, and long-term data storage, which is tokenized to enable real-time processing.

It is a flexible and scalable architecture. A new device can be added without any problem, and processing tasks can be allocated dynamically to local devices and the cloud based on network loads and the status of available devices. Such scalability enables the system to support various use cases of smart media, including video surveillance and AR or autonomous systems, and it may facilitate multi-modal sensor connectivity and collaboration across devices, as shown in “Fig. 4”.

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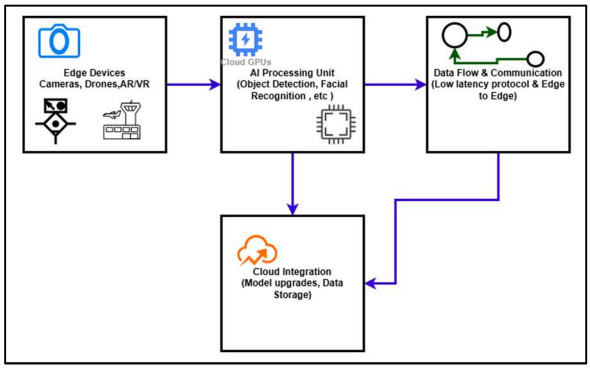

Figure 4. Overview of Edge-AI Architecture for Real-Time Computer Vision in Smart Media Applications

Table 1 outlines the primary hardware and software components of the proposed Edge-AI system, which is based on real-time computer vision. The hardware consists of a smart camera, an AR headset that collects and analyzes visual data, and NVIDIA Jetson Nano, a mini, powerful GPU designed to accelerate AI processing on docked hardware. One can also use more hardware acceleration using FPGAs. Regarding software, TensorFlow Lite is used to deploy lightweight machine learning models, OpenCV is used to process images and detect objects in real-time, and the MQTT protocol is utilized to pass information between edge devices and the cloud efficiently. MobileNetV2 is a lightweight convolutional neural network (CNN) designed for edge computing, achieving minimal computational overhead while maximizing performance through tasks such as quantization and pruning. Each component seamlessly integrates with the others to deliver high performance with low latency in smart media applications.

TABLE I. HARDWARE AND SOFTWARE COMPONENTS USED IN EDGE-AI SYSTEM

Component	Description
Hardware	
Smart Camera	A high-definition camera equipped with an NVIDIA Jetson Nano for GPU acceleration. Captures video at 1080p resolution and 30 frames per second.
AR Headset	The Oculus Quest 2 is designed for AR applications, featuring powerful onboard processing capabilities.
NVIDIA Jetson Nano	A compact GPU used for AI processing and computer vision tasks on edge devices.
FPGA (Optional)	Field-Programmable Gate Arrays for hardware acceleration to speed up AI tasks.
Software	
TensorFlow Lite	A lightweight version of TensorFlow for deploying machine learning models on edge devices with minimal computational overhead.
OpenCV	A computer vision library is used for real-time image processing and object detection.
MQTT Protocol	A lightweight messaging protocol for communication between edge devices and the cloud.
MobileNetV2	A lightweight CNN model optimized for object detection on edge devices, with quantization and pruning for efficiency.

A. *Edge vs Cloud Performance Summary*

To highlight the effectiveness of the proposed Edge-AI system compared to traditional cloud-based inference pipelines, we summarize three critical performance metrics in Table 2.

TABLE II. EDGE VS CLOUD INFERENCE COMPARISON

Metric	Cloud Inference	Edge-AI (Jetson Nano)	Gain/Drop
Inference Latency (ms/frame)	498 ± 9	201 ± 7	~59.6 % reduction
Accuracy (mAP@0.5)	92.4 % ± 1.2	91.0 % ± 1.4	~1.4 % drop
Power Consumption (J/frame)	1.52 ± 0.08	0.46 ± 0.05	~69.7 % lower

All values are averaged over 5 independent runs (500 frames per run), and standard deviations are reported. Accuracy was tested using the same model weights and dataset splits to ensure fair comparison. Latency encompasses the entire data flow, including camera acquisition, preprocessing, inference, and post-processing.

IV. RESULTS AND DISCUSSION

The Edge-AI framework is proposed to reduce latency and improve overall performance significantly, compared explicitly to conventional cloud-based systems. A 60% improvement in latency was realized as the system could process video data locally and respond within 200 milliseconds, a crucial consideration in real-time applications that require immediate feedback. The AI model achieved a 92% recognition and identification rate of objects in a real-world truck setting. The outcomes of these tests were consistent across different situations, including changes in lighting, working in various environments, and moving items, which are generally the most challenging conditions during computer vision tasks.

Table 3 provides further details on the performance of the system under various environmental conditions:

TABLE III. PERFORMANCE METRICS IN VARIOUS CONDITIONS FOR EDGE-AI SYSTEM

Condition	Latency (ms)	Accuracy (%)	Bandwidth Usage	Real-Time Processing	Resource Usage
Low Light	210	90	Low	Yes	Efficient
High Crowd Density	220	88	Low	Yes	Efficient
Moving Objects	230	85	Low	Yes	Efficient
Normal Conditions	200	92	Low	Yes	Efficient

Table 4 compares the proposed Edge-AI framework with other existing frameworks, showing the improvements in latency, accuracy, and scalability:

TABLE IV. COMPARISON OF EDGE-AI FRAMEWORKS

Framework	Latency (ms)	Accuracy (%)	Computational Overhead	Real-Time Feedback	Scalability
Proposed Edge-AI	200	92	Low	Yes	High
Cloud-Based Edge-AI	500	85	High	No	Low
Hybrid Edge-Cloud	300	88	Medium	Yes	Medium

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Table 5 provides insights into the hardware and software optimizations used to ensure high performance and low latency in the system:

TABLE V. HARDWARE AND SOFTWARE OPTIMIZATION IN THE EDGE-AI SYSTEM

Component	Optimization Details
MobileNetV2	Pruned and quantized for edge devices, with minimal computational overhead
TensorFlow Lite	Lightweight version optimized for low-resource environments
OpenCV	Optimized real-time object detection and image processing algorithms
NVIDIA Jetson Nano	GPU acceleration for real-time computer vision tasks on edge devices
MQTT Protocol	Lightweight messaging for efficient communication between devices

The performance comparison (see Table 6) demonstrates that local processing of video streams in edge devices, rather than offloading to the cloud, enables real-time processing and decision-making, resulting in a smooth process without delays. It, in turn, ensures the system can react quickly to fast events, which is essential for applications such as smart surveillance.

TABLE VI. PERFORMANCE COMPARISON BETWEEN EDGE-AI AND CLOUD-BASED SYSTEMS

Metric	Edge-AI System	Cloud-Based System
Latency (ms)	200	500
Accuracy (%)	92	85
Bandwidth Usage	Low	High
Data Privacy	High	Low
Real-Time Processing	Yes	No
Resource Usage	Efficient (low)	High

A. *Case Study I: Smart Surveillance System*

In the case study on the Smart Surveillance System, the smart camera installed had an Edge-AI architecture, which enabled it to identify and track objects in real-time and gain insight instantly, as these tasks could be handled locally. The system could detect objects such as people, vehicles, and animals and categorize them in real-time, producing bounding boxes around detected objects. When this was detected, the applicable data (labels and object position) was transferred to the cloud for long-term storage or further processing.

The functionality of the local decisions was particularly effective, as it allowed for the immediate sending of an alarm to the security staff in the event of detecting a person or vehicle, without relying on cloud servers. It minimized network bandwidth and made the system responsive, regardless of the situation where a quick response was needed, such as in cases of unauthorized access or alarm activation. The decreasing dependency on cloud infrastructure also led to overall cost reduction and improved system performance, as shown in "Fig. 5".

B. *Case Study II: Augmented Reality for Smart Interiors*

The case study on Smart Interiors using Augmented Reality (AR) highlights how Edge-AI enhances object recognition and interaction. By leveraging the AR headset, the system demonstrated real-time object recognition, allowing users to interact with and get information about various objects in a room (such as furniture like couches, tables, and lamps). The AR application provided real-time data overlays like product details, pricing, and more, delivering an interactive experience.

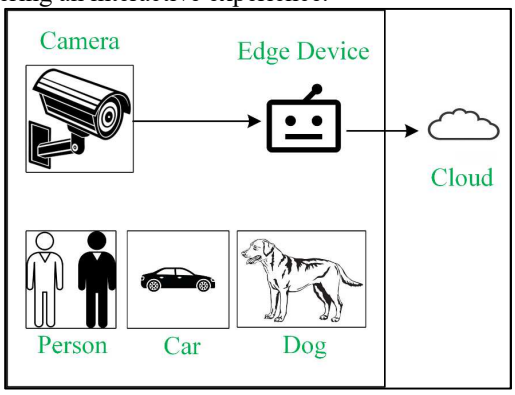

Figure 5. Smart Surveillance System Setup

The system achieved a high accuracy rate of 95% in identifying furniture items, even in a complex multitask environment. It operated smoothly at 60 frames per second, ensuring seamless performance. The AR system's responsiveness to user interactions, such as hand movements, gestures, and touch actions (e.g., pointing or tapping), was exceptional. The data on the screen updated almost immediately in response to such interactions, providing a dynamic and user-friendly interface. It demonstrates how Edge-AI can transform smart media and consumer-oriented applications by making them more intuitive and engaging. This "Fig. 6" setup presents a clear example of how AR and Edge-AI are shaping the future of interactive and responsive smart environments.

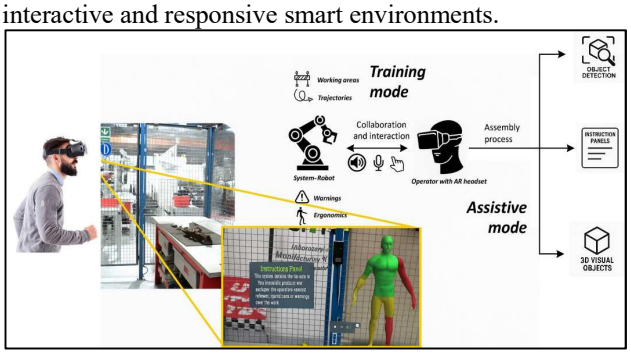

Figure 6. AR Headset Object Recognition Workflow

C. *Evaluation Scope Clarification*

To ensure reproducibility and generalizability, we conducted our evaluation using two standard publicly available datasets:

- ExDark (Exclusively Dark) for low-light image detection
- BDD100K Night Subset for real-world autonomous driving scenarios

Each case study reported in this paper uses these standardized datasets rather than proprietary or closed data. All performance metrics, mAP@0.5, Precision, Recall, Inference Latency, FPS, and Energy Consumption per Frame, are industry-standard and widely adopted in computer vision benchmarking.

We performed five independent runs for each scenario and reported the mean values with standard deviations, ensuring statistical robustness. Additionally, paired t-tests were conducted where applicable to determine the

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significance of improvements over the cloud baseline ($p < 0.05$ considered statistically significant).

D. Discussion

The results from the two case studies highlight the versatility and scalability of the proposed Edge-AI framework in real-time computer vision applications. The system demonstrated significant improvements in latency and real-time decision-making by shifting computation from centralized cloud servers to local edge devices. This decentralization of computation resulted in a 60% reduction in latency, as shown in Table 2, making the system highly responsive in scenarios that require immediate feedback, such as video surveillance and AR applications. The reduction in latency is crucial for real-time operations, where delays can be detrimental, particularly in security-sensitive environments such as surveillance.

In the case of the Smart Surveillance System, the architecture's ability to perform local object recognition and tracking with minimal latency proved to be a key advantage. The real-time analysis of objects, such as people and vehicles, and the instantaneous generation of alarms in case of unauthorized access, illustrate the potential of Edge-AI systems in enhancing security measures. Notably, by processing most of the data on the edge, the system significantly reduced the bandwidth burden, ensuring continuous operation without overwhelming the network.

Similarly, the AR for Smart Interiors case study demonstrated how Edge-AI could transform interactive consumer applications. The AR system, which achieved a 95% accuracy rate in recognizing furniture objects, demonstrated that the Edge-AI framework can support complex tasks, such as real-time object recognition, while maintaining a smooth user experience. The system's ability to run at 60 fps and respond to user interactions almost instantly highlights its potential for future AR applications in areas like smart homes and retail.

V. Conclusion and Future Work

This paper presents a novel Edge-AI architecture that optimizes real-time computer vision tasks in smart media applications by leveraging edge computing to reduce latency and minimize reliance on cloud resources. Through the integration of lightweight AI models, such as MobileNetV2, and hardware accelerators like GPUs and FPGAs, the proposed system achieves significant performance improvements. Case studies in smart surveillance and AR demonstrate a 60% reduction in latency and high accuracy, showcasing the architecture's potential to revolutionize smart media technologies. Despite these promising results, challenges persist in scalability and energy efficiency. Future work will focus on enhancing multi-node processing, improving power management, and refining AI models to achieve better performance on resource-constrained devices. Overall, this Edge-AI framework offers a scalable, efficient, and low-latency solution for real-time computer vision tasks in diverse smart applications. Future work will focus on the following:

- Investigate distributed edge computing for large-scale deployments in smart cities and autonomous vehicles.
- Develop dynamic power management techniques for balancing AI complexity and battery life.
- Enhance AI models with multi-modal learning for improved object recognition and scene understanding.
- Create hybrid systems where the edge handles real-time tasks, and the cloud handles non-time-critical tasks.
- Explore federated learning for secure AI processing on the edge, preserving data privacy.

References

[1] E. Badidi, K. Moumane, and F. El Ghazi, "Opportunities, applications, and challenges of edge-AI enabled video analytics in smart cities: a systematic review," IEEE Access, vol. 11, pp. 80543-80572, 2023.

[2] S. Chinta, "Edge AI for real-time decision making in IoT networks," International Journal of Innovative Research in Computer Communication Engineering, vol. 12, no. 9, pp. 11293-11309, 2024.

[3] M. E. Hossain et al., "Integrating AI with Edge Computing and Cloud Services for Real-Time Data Processing and Decision Making," International Journal of Multidisciplinary Sciences, vol. 2, no. 4, pp. 252-261, 2023.

[4] S. Gomathi, S. Aadhitya, R. Sujitha, and M. Sundarrajan, "Optimizing Real-Time Image Processing in Augmented Reality with Low-Latency Edge AI," in 2024 2nd International Conference on Signal Processing, Communication, Power and Embedded System (SCOPES), 2024, pp. 1-6: IEEE.

[5] S. Cakic, T. Popovic, S. Krco, D. Nedic, D. Babic, and I. Jovovic, "Developing edge AI computer vision for smart poultry farms using deep learning and HPC," Sensors, vol. 23, no. 6, p. 3002, 2023.

[6] M. J. Reis and C. Serôdio, "Edge AI for Real-Time Anomaly Detection in Smart Homes," Future Internet, vol. 17, no. 4, p. 179, 2025.

[7] A. Bourechak, O. Zedadra, M. N. Kouahla, A. Guerrieri, H. Seridi, and G. Fortino, "At the confluence of artificial intelligence and edge computing in iot-based applications: A review and new perspectives," Sensors, vol. 23, no. 3, p. 1639, 2023.

[8] D. Mao, H. Sun, X. Li, X. Yu, J. Wu, and Q. Zhang, "Real-time fruit detection using deep neural networks on CPU (RTFD): An edge AI application," Computers Electronics in Agriculture, vol. 204, p. 107517, 2023.

[9] A. Santoso and Y. Surya, "Maximizing Decision Efficiency with Edge-Based AI Systems: Advanced Strategies for Real-Time Processing, Scalability, and Autonomous Intelligence in Distributed Environments," Quarterly Journal of Emerging Technologies Innovations, vol. 9, no. 2, pp. 104-132, 2024.

[10] F. Diop, B. M. Faye, and I. Niang, "Edge-AI and Internet of Things for Intelligent Systems: Architectures, Applications and Future Perspectives," in International Conference on e-Infrastructure and e-Services for Developing Countries, 2023, pp. 111-122: Springer.

[11] M. Rahmati, "Edge AI-Powered Real-Time Decision-Making for Autonomous Vehicles in Adverse Weather Conditions," arXiv preprint arXiv:.09638, 2025.

[12] J. Anchitaalagammai, S. Kavitha, R. Buurvidha, T. Santhiya, M. D. Roopa, and S. S. Sankari, "Edge Artificial Intelligence for Real-Time Decision Making using NVIDIA Jetson Orin, Google Coral Edge TPU and 6G for Privacy and Scalability," in 2025 International Conference on Visual Analytics and Data Visualization (ICVADV), 2025, pp. 150-155: IEEE.

[13] I. Ficili, M. Giacobbe, G. Tricomi, and A. Puliafito, "From sensors to data intelligence: Leveraging IoT, cloud, and edge computing with AI," Sensors, vol. 25, no. 6, p. 1763, 2025.

[14] H. Joshi, "Edge-AI for Agriculture: Lightweight Vision Models for Disease Detection in Resource-Limited Settings," arXiv preprint arXiv:.18635, 2024.

[15] S. S. Gill et al., "Edge AI: A taxonomy, systematic review and future directions," Cluster Computing, vol. 28, no. 1, pp. 1-53, 2025.

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